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A Deep Learning Framework for Hyperspectral Image Classification using PCA and Spectral LSTM Networks

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***Abstract*- Deep learning techniques, like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have demonstrated remarkable performance in hyperspectral image (HSI) classification in recent years. This paper proposes a novel framework for the classification of hyperspectral images using a Long Short-Term Memory Networks (LSTM) and Principal Component Analysis (PCA). PCA is basically used to minimize the high-dimensional image data, and then it send to LSTM for classification. LSTM is a RNN that is designed to perform better on sequential analysis of hyperspectral images. This proposed research and analysis of HIS data was performed on the standardized dataset of Pavia University and Kennedy Space Center (KSC). The proposed research framework effectively classifies the HIS images and shows its efficency in comparison to existing state-of-the-art classifiers. Overall, the proposed framework presents a promising approach to the classification of hyperspectral images by combining the strengths of LSTM and PCA. The paper contributes to the literature on hyperspectral image classification by demonstrating the effectiveness of deep learning models and the importance of considering the sequential nature of hyperspectral data.**

Keywords-
Hyperspectral Imaging, Long Short-Term Memory Networks (LSTM), Principal Component Analysis (PCA),Image Classification; Deep Learning.

I. INTRODUCTION

Hyperspectral sensors can capture numerous closely spaced spectral bands across a wide range of wavelengths, extending from the visible to the near-infrared spectrum.This high spectral resolution allows for the detection and identification of subtle differences in the reflectance and absorption properties of materials on the Earth's surface.By analyzing the spectral signatures of these materials, hyperspectral data can be used to identify and map different land covers, vegetation species, minerals, and other features on the Earth's surface. HIS can also be used for different types of applications domains, such as agriculture, mineral exploration, environmental monitoring, and disaster management. Each pixel in a hyperspectral image (HSI) can be thought of as a high-dimensional feature vector whose elements correspond to spectral reflectance at a certain wavelength. HSIs have been extensively used in several fields due to their ability to detect small spectral differences. The most active subject of research in the hyperspectral community is the classification of HS imaging. It means assigning each pixel to a certain class based on its unique spectral properties which has gotten a lot of interest in the remote sensing sector. There are two key obstacles with HSI classification tasks: 1) the spectral signals' significant geographical diversity, and 2) the little training available samples vs. hyperspectral data's high dimensionality. Many elements, such as variations in illumination, ambient, climatic, and temporal circumstances, contribute to the first challenge. The second difficulty will provide indistinct worries for a number of approaches and impair the capability of classifiers to generalize [1-2]. The field of hyperspectral imagery has recently progressed due to the rapid development of optics and photonics. As a result, several satellites now include hyperspectral sensors that can create photos with a lot of spectral information. Using satellites, it is easy to differentiate between extremely similar materials and objects with the help of extensive information obtained in hyperspectral photos. A small reconnaissance imaging spectrometer for Mars hyperspectral imaging dataset [3] was examined, and subsequently the mineralogy of the Martian surface was determined using linear mixing of absorption band approaches. The apparatus employed for mineral exploration in [4] utilized a hyperspectral scanning push broom device designed for visible and near-infrared (VNIR) imaging spectroscopy. This device was capable of capturing data within the VNIR spectrum, spanning wavelengths from 400 to 1000 nm.. To simultaneously tackle the challenges of label scarcity and high-dimensional data in hyperspectral image classification, the initial step involves exploring the interrelationships among different pixels. The classification process in hyperspectral pictures can benefit from spatial layout information [6-7]. Post processing is used to combine the findings of numerous classification/segmentation procedures to create the final spectral-spatial classification map. The geographical or contextual information is not directly incorporated into the classifier in most of the aforementioned HSI picture categorization approaches. Many computers vision tasks have been proposed using sparse representation where the use of sparsity as a prior often leads to better results [8-9]. The use of different types of techniques used for feature reduction,feature extraction and selection, such as Principal component analysis (PCA) [9-10][18] and linear discriminant analysis (LDA) [11] are two examples of this type of study. Based on the classification process, hyperspectral image classification

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areof three categories: supervised classification [12-13], Semi-Supervised [14-16] and unsupervised classification [17].In supervised classification, pixels are assigned labels based on the least distance constraint, utilizing the combined sparse coefficients and structured dictionary for the ultimate classification. Experimental results on two authentic hyperspectral image datasets validate the effectiveness of this approach[10]. A Kernel ELM for the supervised classification of hyperspectral images has been proposed which produced satisfactory results [13]. Reference [17] provides models for the fully automatic and unsupervised solution of clustering, feature identification, and class number estimation issues at the same time.

II . BACKGROUND OF THE PROPOSED WORK

A. Datasets

Our research and analysis were carried out using the standard datasets from Kennedy Space Center (KSC) and Pavia University. These hyperspectral image datasets consist of multiple spectral bands and provide descriptions of various geographical landscapes categorized into different classes. Each class specify a unique kind of data present in that area. To view the complete dataset, all bands or certain no. of bands are to be stacked upon one another to visualise the data. Certain classes of data can only be seen after stacking some specific set of bands together which might be different for some different set of bands. During classification, some bands do not provide any relevant data information due to which they are discarded from the dataset.

B. Pavia University

The Pavia University dataset is a hyperspectral image dataset that was collected over the Pavia region in northern Italy using the Reflective Optics System Imaging Spectrometer (ROSIS) technique. The dataset contains hundred three spectral bands, with a total image resolution of 610 x 610 x 103 pixels. The ground truth information for this dataset is classified into nine different classes. Hyperspectral images like this one contain information about the spectral characteristics of each pixel in the image, which can be used to identify different materials or objects in the scene. In this case, the nine classes in the ground truth information likely correspond to different land cover types, such as buildings, vegetation, water, etc. The high spectral resolution of the dataset means that it may be possible to distinguish between different types of vegetation or other materials that might appear similar in other types of images. However, the large number of bands also means that processing and analyzing the data can be computationally intensive.

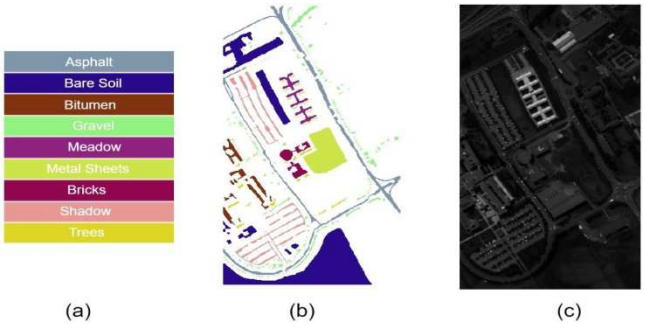


Fig.1 Pavia University dataset

C. Kennedy Space Center

The KSC dataset is a hyperspectral image dataset collected using the NASA AVIRIS instrument at the Kennedy Space Center in Florida. The dataset contains 224 spectral bands, and the environmental land coverings have been classified into 13 different classes with a total image resolution of the final image is 512 x 614 x 176 pixels. Hyperspectral images like this one provide a wealth of information about the spectral characteristics of the scene being imaged. This information can be used to identify different materials or objects in the scene, and to study the composition and structure of the environment. The 13 classes in the ground truth information likely correspond to different land cover types, such as vegetation, water, urban areas, and bare ground. The high spectral resolution of the dataset means that it may be possible to distinguish between different types of vegetation or other materials that might appear similar in other types of images. Like Pavia University dataset, the large number of bands in the Kennedy Space Center dataset can make processing and analyzing the data computationally intensive.

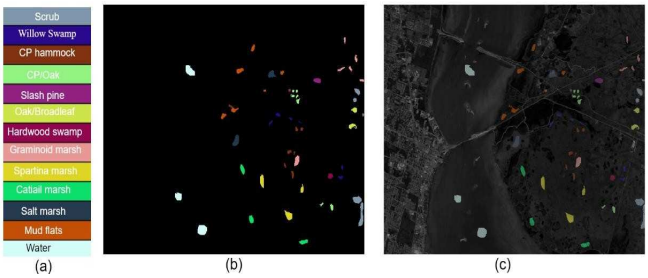


Fig.2 Kennedy Space Center dataset

For the execution of the project, the image file along with its ground truth is provided to the proposed model and the data are stored in matrix format. The shape of the matrix is detected which will be used for future reference. Further, the pixels of the image are extracted and stored in a CSV format. The CSV file helps to gather information about the dataset. This includes the type of data, the no of rows and columns, memory usage, etc. The dataset details are demonstrated in Table I.

Table I Dataset Details Pavia University, Kennedy Space Center

Name of Datasets	No. of Class	No.of Bands	No. of Reduced Band[a]
Pavia University	9	103	103
Kennedy Space Center	13	224	176

D. Principal Component Analysis

Principal Component Analysis (PCA) is used for reducing the dimensionality of high-dimensional data while preserving most important features of the given dataset. PCA works by transforming the given data into a new coordinate system that is defined by the principal components(PC) of the data. These PC are linear collection of the original variables that are un-correlated and capture the highest amount of variance in the data. The first principal component is the direction in which the data varies the most, and each subsequent component captures as much of the remaining variation as possible. By keeping only the top k principal components, PCA effectively reduces the

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dimensionality of the data from n dimensions to k dimensions, where k << n.

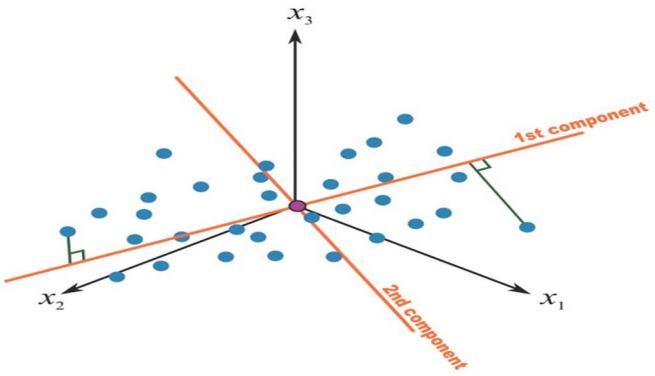


Fig.3 Principal Component Analysis (PCA)

E. Long Short-Term Memory Networks

LSTM has a chain formation that restrains four neural structures and several memory blocks known as cells as depicted in Figure 4.

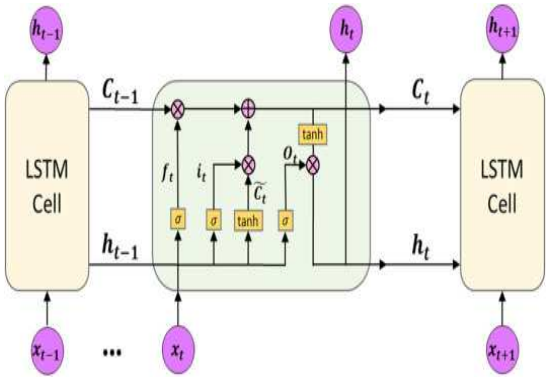


Fig.4 LSTM chain structure

LSTMs are a type of RNN(recurrent neural network) that are designed to capture long-term dependencies in sequential data. The key innovation of LSTMs is the use of memory cells and gates to control the flow of information through the network. Each memory cell in an LSTM network retains information over time and can selectively update or forget this information based on input from the gates. The forget gate is responsible for deciding which information should be removed from the cell, whereas the input gate dictates which new information should be incorporated into the cell. The output gate controls the flow of information from the cell to the output of the network. Finally, the use gate (also called the candidate gate) determines which information should be used from the cell to update the output. [18,19]. Mathematically, an LSTM structure evaluates a mapping using an input sequence .After the forget gate determines the degree of information to remember or discard, The LSTM cell proceeds to calculate the new information that should be stored in the cell state, considering the input xit and the previous cell state $Ct-1$. It then produces ft, which represents the information of the degree to be retained. ft value will be presented as:

$$ft = \sigma(Wf.\,[Hit-1, xit] + bf) \quad (1)$$

where $Hit-1$ is the output value from the last state.

Following this step, the next crucial task is to identify the new information that should be stored within the cell, thus forming the new cell state. This can be done in two primary steps:

1. A sigmoid hidden layer function, also known as the input- gate layer, determines which values from the input and previous hidden state will be selected for updating the cell state. This is computed using the equation

$$lt = \sigma(Wl.\,[Hit-1, xit + bi]) \quad (2)$$

where $[Hit-1, xit]$ is the concatenation of the previous hidden state and the current input.

2. A tanh hidden layer function forms a vector of new candidate values $\tilde{C}$ that are selected to add in further states. This is computed using the equation.

$$\tilde{C}=tanh(WC[Hit-1,xit]+bC) \quad (3)$$

where $[Hit-1, xit]$ is the concatenation of the previous hidden state and the current input.

Now

The updated information is then integrated to the old cell state $Ct-1$ to provide the new cell state Ct using element-wise multiplication (Hadamard product) of the forget gate output ft and the old cell state $Ct-1$, and element-wise multiplication of the input gate output lt and the candidate values $\tilde{C}t$.with ft and adding

$$lt \times \tilde{C}: \quad = ft \circ Ct-1 + lt \circ \tilde{C}t \quad (4)$$

Finally, the output is computed by passing the updated cell state Ct through a tanh activation function and multiplying it by the output gate output Ot, which is computed using the equation:

$$Oit = ([hit-1, xit] + boutput \quad (5)$$

where $[hit-1, xit]$ is the concatenation of the previous hidden state and the current input,

and $Woutp$ is the output gate weight matrix

The final output of the LSTM cell is the updated hidden state, computed using the equation

$$hit = Ot \times tanh(Ct) \quad (6)$$

III. METHODOLOGY

The proposed PCA-LSTM framework is designed to process hyperspectral images by first applying PCA to reduce the dimensionality of the spectral data, and then using LSTM (Long Short-Term Memory) neural networks to analyze the resulting reduced dataset. PCA (Principal Component Analysis) is a common technique used to reduce the dimensionality of high-dimensional data. In this case, PCA is applied to the hyperspectral images to reduce the number of spectral bands and extract the most important spectral features. The goal is to preserve as much spatial information as possible while maximizing the variance in the reduced dataset.[20][21]The resulting minimize dataset is then selected for training, validation, and testing data sets for use in the LSTM neural network. LSTM is a type of recurrent neural network that is well-suited for processing sequential data, such as time-series or sequential image data. By combining PCA and LSTM, the proposed framework aims to improve the accuracy of hyperspectral image classification by reducing the dimensionality of the spectral data while preserving important spatial information, and then using LSTM to analyze the resulting reduced dataset. This approach has the potential to be more computationally efficient than traditional spectral-spatial frameworks that process the entire hyperspectral cube, while still achieving high classification accuracy.[22] The proposed PCA-LSTM

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framework is as follows

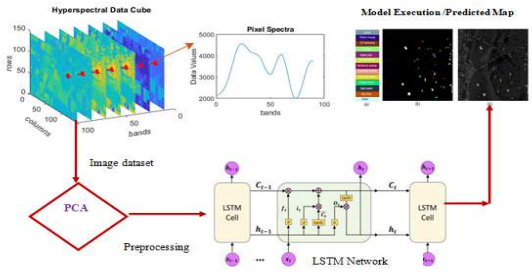


Fig. 5 Proposed Framework

As shown in fig-5 once the training data has been processed through PCA and through preprocessing stage the data set divided into training, validation, and testing sets and it is passed to the LSTM neural network for training. LSTM is a type of recurrent neural network that is designed to process sequential data by learning to remember relevant information over time. In this case, the LSTM network is trained on the reduced spectral dataset to learn the relationships between the spectral features and the corresponding land cover classes. During training, the LSTM network learns to remember the essential sequences of the input data by using a forget gate mechanism to selectively retain or discard information from previous time steps. This enables the network to build meaningful connections between the input sequences and the corresponding land cover classes. Once the LSTM network has been trained, a classifier is obtained that can be used to predict the land cover classes of new hyperspectral images. After the testing data is fed into the trained classifier, it proceeds to predict the class of each pixel within the input image. These obtained classes for prediction are subsequently matched to the ground- truth information to assess the model's performance.[21]

IV . RESULTS AND ANALYSIS

PCA-LSTM is a framework used for temporal modeling of hyperspectral data. Hyperspectral images (HSIs) are captured over time and consist of multiple spectral bands. LSTM networks can capture temporal dependencies in these bands by processing them sequentially. The framework uses PCA to reduce the dimensionality of the spectral data while preserving important spatial information. The reduced dataset is divided into training, validation, and testing sets. In this study, a small percentage (5%) of the data is used for training and another 5% for validation, while the remaining 90% is used for testing. The model undergoes evaluation using a range of metrics, including overall accuracy, producer's accuracy, user's accuracy, and the kappa coefficient. The experimental findings conclusively show that the PCA-LSTM framework we propose attains exceptional classification accuracy for both the Kennedy Space Center and Pavia University datasets, surpassing several established state-of-the-art techniques. This underscores the potential of the PCA and LSTM fusion as a highly promising method for hyperspectral image classification and its adaptability to diverse remote sensing applications.

This suggests that the combination of PCA and LSTM is a promising approach for hyperspectral image classification, which could be extended to other remote sensing applications.[12][22]

The evaluation of the proposed framework involves comparing the predicted values with the actual ground truth values using various metrics such as overall accuracy, producer's accuracy, user's accuracy, and kappa coefficient. Overall accuracy is the percentage of correctly classified pixels out of the total number of pixels. Producer's accuracy is the proportion of correctly classified pixels in each class to the total number of pixels in that class. User's accuracy is the proportion of correctly classified pixels in each class to the total number of pixels predicted to belong to that class. Kappa coefficient is a statistical measure that compares the agreement between predicted and actual classifications, taking into account the level of agreement expected by chance. The higher the values of these metrics, the better the classification accuracy of the model.

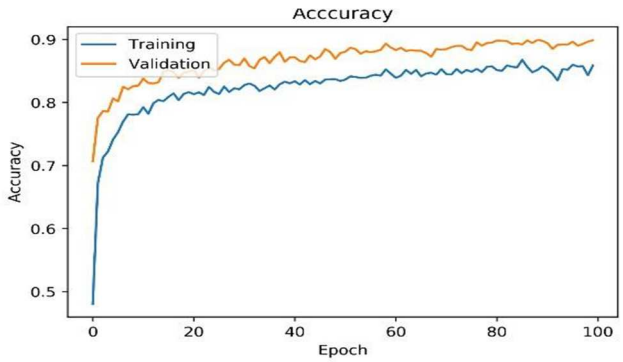


Fig.6 Accuracy curves of LSTM in case of KSC Dataset

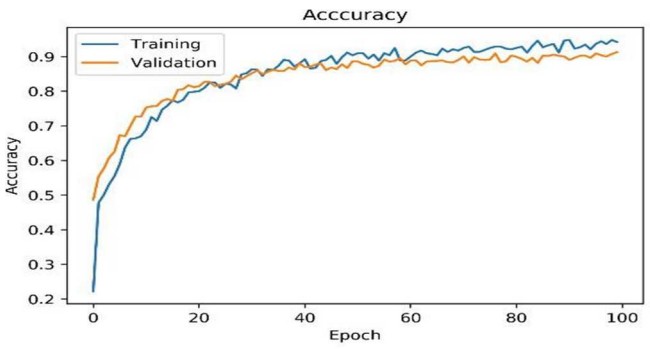


Fig.7 Accuracy curves of LSTM in case of Pavia University Dataset

Table II- Comparison of the Proposed framework of LSTM with PCA and other models

SL. No	Dataset	LSTM +PCA	SAE	EN	KNN	SVM
1	KSC	90.00	65.11	66.24	67.70	78.14
2	Pavia	90.00	62.11	59.73	61.19	63.89

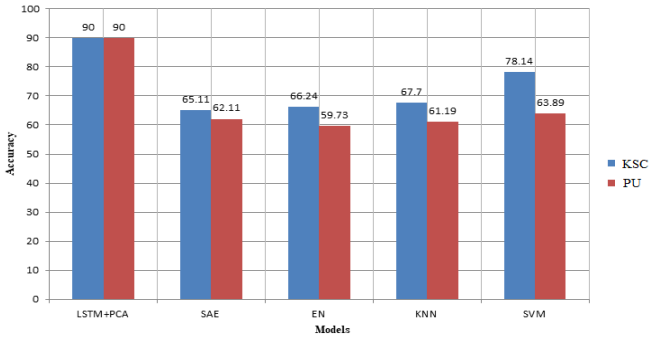
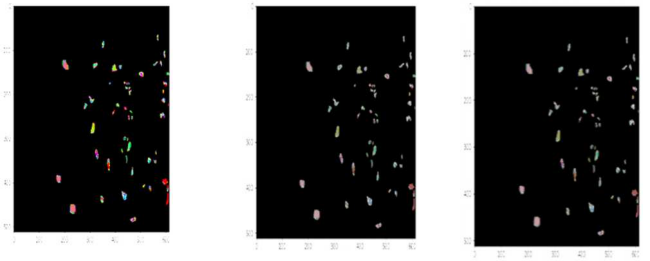
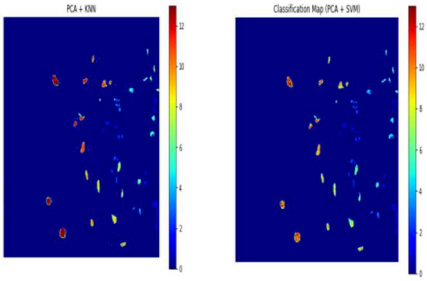


Fig.8 Accuracy Comparison of the Proposed framework of LSTM

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(a) Proposed Framework (b) Classification map using SAE (c) Classification map using EN



(d) Classification map using KNN (e) Classification map using SVM

Fig.9 Classification map (KSC) of the (a) Proposed Framework (b) SAE (c) EN (d) KNN (e) SVM with PCA and other models of KSC and PU

The PCA-LSTM framework combines PCA (Principal Component Analysis) with LSTM for HSI classification. Proposed method compared with some deep learning models named Stacked Auto encoders-Deep Networks (SAE), Ensemble-Boosted (EN) and traditional machine learning algorithms such as The proposed framework demonstrated superior performance in comparison to all other models, including K-Nearest Neighbour (KNN) and Support Vector Machines (SVM), when assessed in terms of overall accuracy (OA), as shown in Table 2 and Evaluation metrics such as overall accuracy, kappa coefficient, and F1-score used to compare the performance of different methods are shown in Table 3 and 4.

The suggested framework useing LSTM is able to remember important information from previous time steps using the forget gate. This is a key advantage of the LSTM model, as it allows the model to retain relevant information and discard irrelevant information, which can be especially important in HSI classification where there may be long-term dependencies and complex temporal dynamics. Overall, the forget gate in LSTM networks is a powerful mechanism that enables the model to better capture the temporal dynamics of the data, and the proposed framework takes advantage of this capability to improve HSI classification performance Classification maps are shown in Figures 9and 10

Table III Precision, Recall, and F1-Score of the Proposed Framework "KRC" for Individual Classes

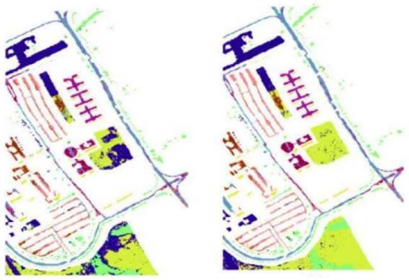
	Kennedy Space Center		
Classes	**Precision**	**Recall**	**f1-score**
1	0.96	0.95	0.95
2	0.86	0.90	0.88
3	0.89	0.95	0.92
4	0.76	0.63	0.69
5	0.61	0.75	0.67
6	0.72	0.63	0.68
7	0.76	0.90	0.82
8	0.94	0.90	0.92
9	0.95	0.97	0.96
10	0.96	0.96	0.96
11	0.99	0.99	0.99
12	0.92	0.95	0.93
13	1.00	0.98	0.99
	Overall Accuracy		90%

Table IV Detailed Analysis of Individual Class Precision, Recall, and F1-Score for the Proposed Framework Pavia University

	Pavia University		
Classes	**Precision**	**Recall**	**f1-score**
1	0.90	0.90	0.90
2	0.94	0.96	0.95
3	0.71	0.73	0.72
4	0.87	0.89	0.88
5	0.99	0.99	0.99
6	0.91	0.83	0.87
7	0.81	0.70	0.75
8	0.80	0.80	0.80
9	0.98	1.00	0.99
	Overall Accuracy		**90%**



(a) Proposed Framework (b) Classification map using SAE (c) Classification map using EN



(d) Classification map using KNN (e) Classification map using SVM

Fig. 10 Classification map of Pavia University dataset of (a) Proposed Framework (b) SAE (c) EN (d) KNN (e) SVM

V. CONCLUSIONS

The proposed PCA-LSTM based hyperspectral classification framework combines the efficiency of PCA

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for dimensionality reduction with the power of LSTM for sequential data analysis. The framework is able to learn the relationships between the spectral features and the corresponding land cover classes with high accuracy, outperforming several existing commonly used methods. The experimental results showcase that the proposed framework enhances accuracy by a remarkable 138% for the Kennedy Space Center (KSC) dataset and an impressive 151% for the Pavia University dataset, respectively.. This indicates that the combination of PCA and LSTM is a promising approach for hyperspectral image classification, which could have significant applications in remote sensing and other related fields. Overall, this work provides a novel and effective method for classification of hyperspectral images, which could pave the way for further research in this area.

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